Change over time in the sibilant space of African American Language in Washington D.C.

Griffin Lowry

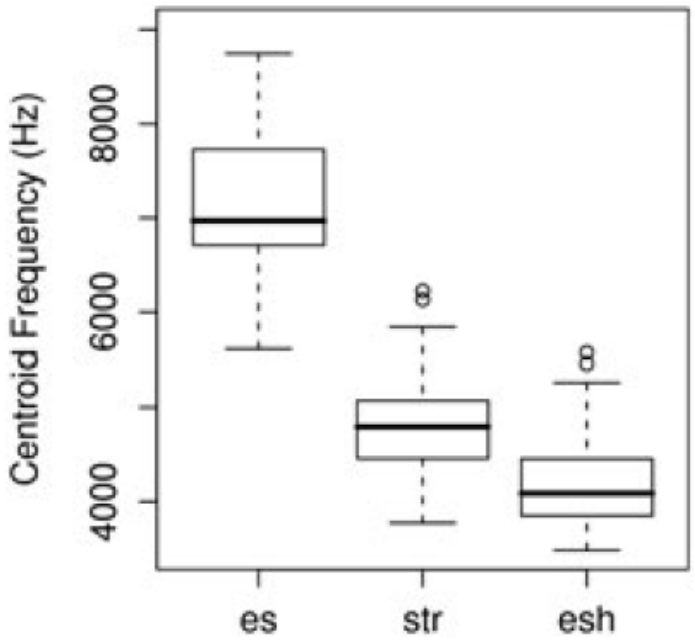
Indiana University (Bloomington)

22 October 2026 - NWAV 54



/stɹ/ retraction

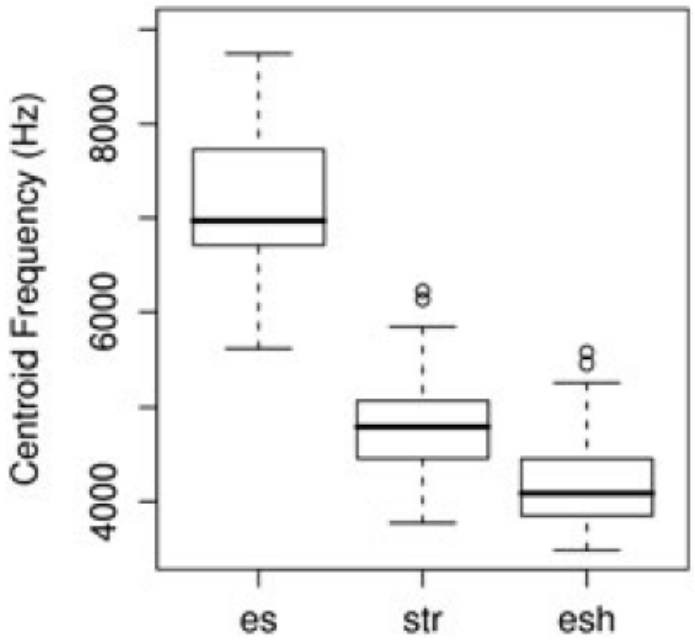
- The retraction of /s/ to [ʃ] in /stɹ/ clusters is a commonly examined sound change (e.g., Baker et al., 2011; Wilbanks, 2016)



Baker et al. (2011)

/stɹ/ retraction

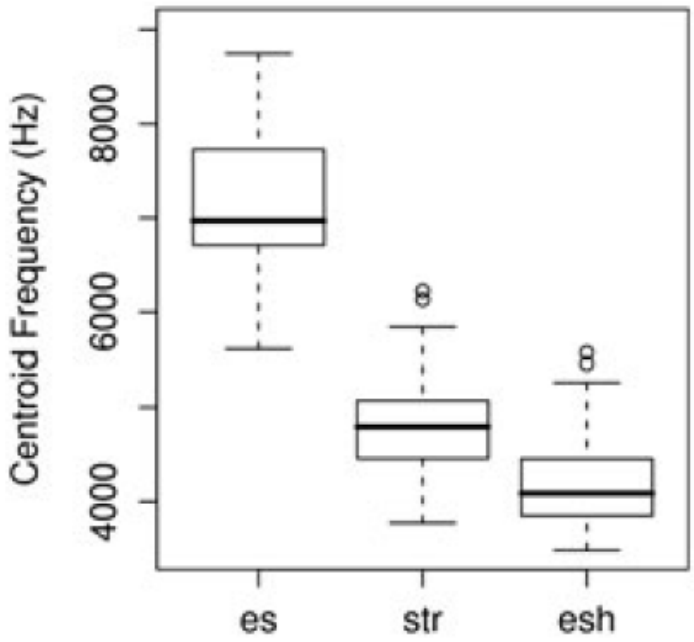
- The retraction of /s/ to [ʃ] in /stɹ/ clusters is a commonly examined sound change (e.g., Baker et al., 2011; Wilbanks, 2016)
  - The tongue retracts backwards, causing a lower frequency



Baker et al. (2011)

/stɹ/ retraction

- The retraction of /s/ to [ʃ] in /stɹ/ clusters is a commonly examined sound change (e.g., Baker et al., 2011; Wilbanks, 2016)
  - The tongue retracts backwards, causing a lower frequency
- Phonetic or phonological?



Baker et al. (2011)

Phonetic bias towards /stɹ/ retraction

- Baker et al. (2011) found that non-retractors still show coarticulatory bias towards retraction

Phonetic bias towards /stɹ/ retraction

- Baker et al. (2011) found that non-retractors still show coarticulatory bias towards retraction
- They classify a retracted /s/ as being ≥75.5% of the acoustic distance from [s] to [ʃ]

Phonetic bias towards /stɹ/ retraction

- Baker et al. (2011) found that non-retractors still show coarticulatory bias towards retraction
- They classify a retracted /s/ as being ≥75.5% of the acoustic distance from [s] to [ʃ]
  - [s]tɹ was 48% of the way towards [ʃ] for non-retractors

/stɹ/ in AAL

- /stɹ/ retraction was found to be at an *advanced* stage in Washington D.C. AAL (Gunter et al., 2021)

/stɹ/ in AAL

- /stɹ/ retraction was found to be at an *advanced* stage in Washington D.C. AAL (Gunter et al., 2021)
- Many speakers retracted /stɹ/ beyond their mean /ʃ/

/stɹ/ in AAL

- /stɹ/ retraction was found to be at an *advanced* stage in Washington D.C. AAL (Gunter et al., 2021)
- Many speakers retracted /stɹ/ beyond their mean /ʃ/
- Men were also retracting /spɹ/ and /skɹ/

/stɹ/ in AAL

- /stɹ/ retraction was found to be at an *advanced* stage in Washington D.C. AAL (Gunter et al., 2021)
- Many speakers retracted /stɹ/ beyond their mean /ʃ/
- Men were also retracting /spɹ/ and /skɹ/
- "/s/ retraction may be indexing a local and older vernacular form in Washington DC AAL."

Change over time with /stɹ/

- Most research on /stɹ/ retraction has not directly examined change over time

Change over time with /stɹ/

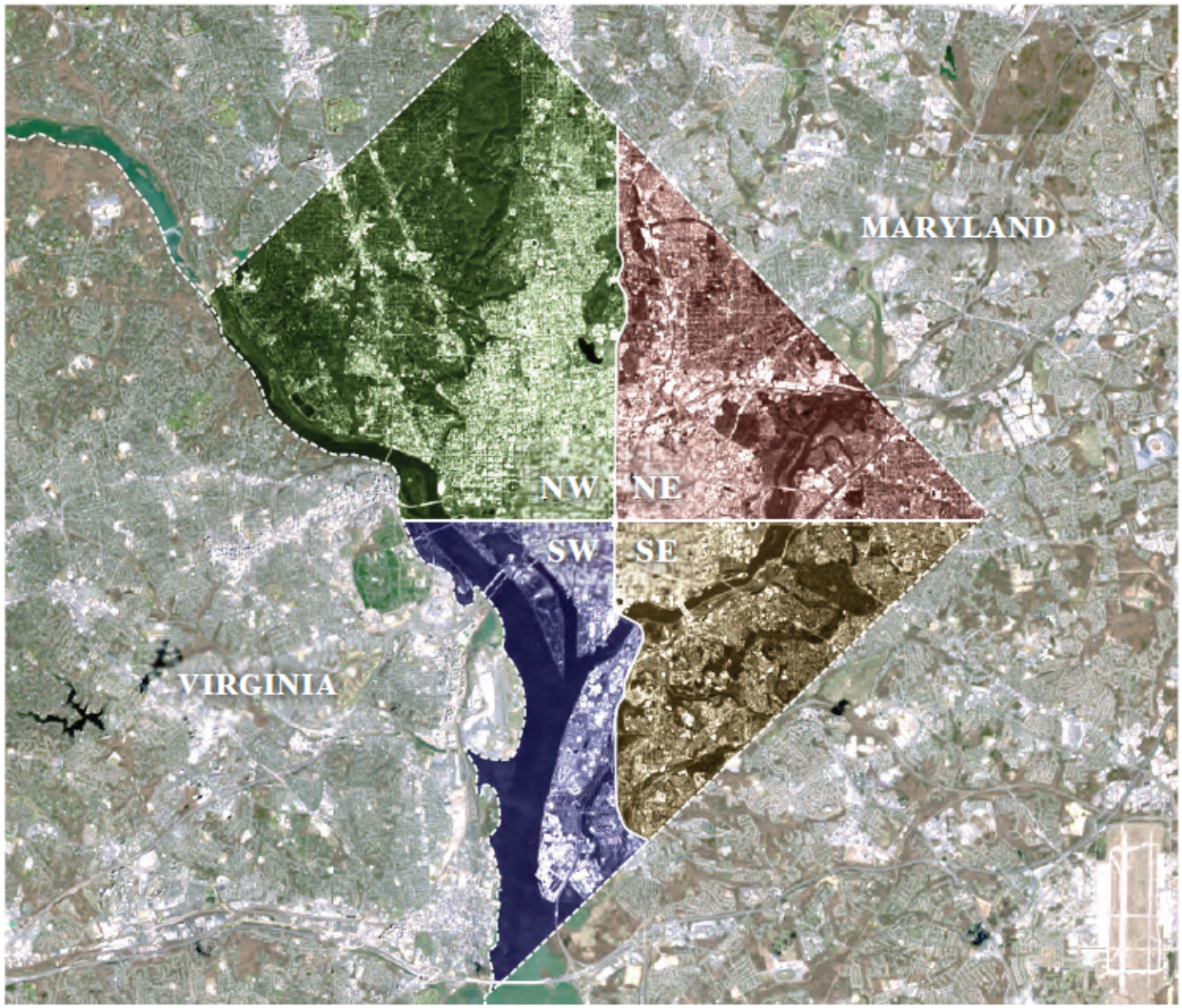
- Most research on /stɹ/ retraction has not directly examined change over time
  - Gunter et al. (2021) did an apparent time analysis

Change over time with /stɹ/

- Most research on /stɹ/ retraction has not directly examined change over time
  - Gunter et al. (2021) did an apparent time analysis
- Opportunity to compare two corpora of Washington D.C. AAL recorded over 45 years apart

FIGURE 2

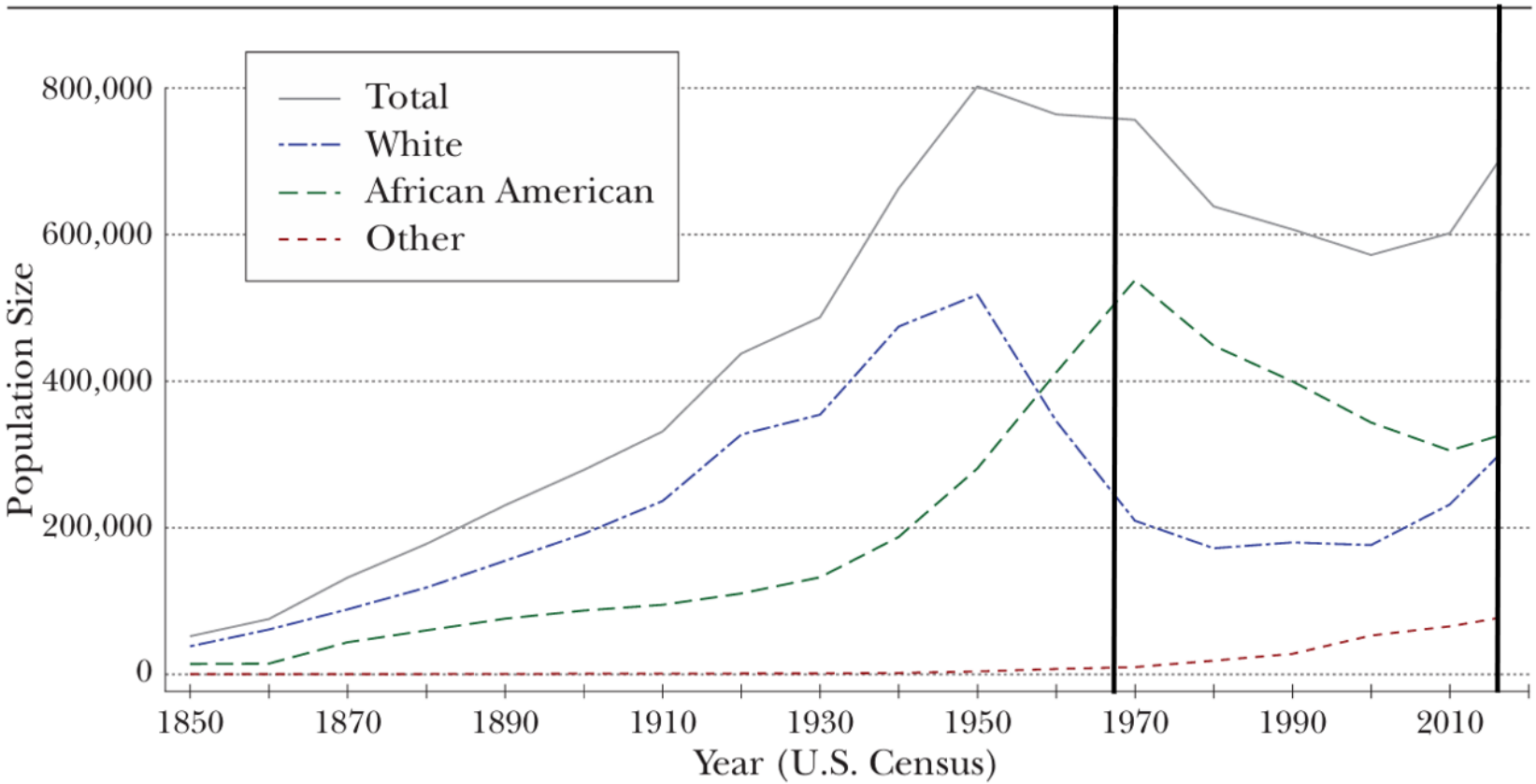
Satellite Image of Washington, D.C., with Quadrants Highlighted



Farrington and Schilling (2019)

FIGURE 1

Population of African Americans, Whites, and Others in Washington, D.C.



Farrington and Schilling (2019)

Research Questions

- Did /stɹ/ retraction increase between 1968 and 2017 in Washington D.C. AAL?
  - Hypothesis: /stɹ/ retraction was less prominent in the 1960s than in the 2010s
- Did coarticulatory bias towards /stɹ/ retraction increase between over time?
  - Hypothesis: as /stɹ/ retraction becomes more phonologized in a community, coarticulatory bias in non-retractors also increases

The Corpus

- DCA (sample)
  - Recorded from 1968-69
  - 23 speakers (7 female)
  - 6087 sibilant tokens
  - 163 /stɹ/ tokens
- DCB (sample)
  - Recorded from 2015-17
  - 43 speakers (24 female)
  - 22,015 sibilant tokens
  - 668 /stɹ/ tokens



Farrington and Kendall (2019)

Data Collection

- Measured all word-initial tokens of /s/ and /ʃ/

Data Collection

- Measured all word-initial tokens of /s/ and /ʃ/
- Automatically collected spectral moment data using `one_script` (Mielke and Wilbanks, 2015)

Data Collection

- Measured all word-initial tokens of /s/ and /ʃ/
- Automatically collected spectral moment data using `one_script` (Mielke and Wilbanks, 2015)
- Created multitaper spectra and measurements in R (Reidy, 2013)

Data Collection

- Measured all word-initial tokens of /s/ and /ʃ/
- Automatically collected spectral moment data using `one_script` (Mielke and Wilbanks, 2015)
- Created multitaper spectra and measurements in R (Reidy, 2013)
  - 8 tapers, no preemphasis

Data Collection

- Measured all word-initial tokens of /s/ and /ʃ/
- Automatically collected spectral moment data using `one_script` (Mielke and Wilbanks, 2015)
- Created multitaper spectra and measurements in R (Reidy, 2013)
  - 8 tapers, no preemphasis
- Excluded tokens below 2400 Hz (Stuart-Smith et al., 2019)

Data Collection

- Measured all word-initial tokens of /s/ and /ʃ/
- Automatically collected spectral moment data using `one_script` (Mielke and Wilbanks, 2015)
- Created multitaper spectra and measurements in R (Reidy, 2013)
  - 8 tapers, no preemphasis
- Excluded tokens below 2400 Hz (Stuart-Smith et al., 2019)
- Excluded speakers with less than 5 tokens of /stɪ/ (Wilbanks, 2016)

Analysis

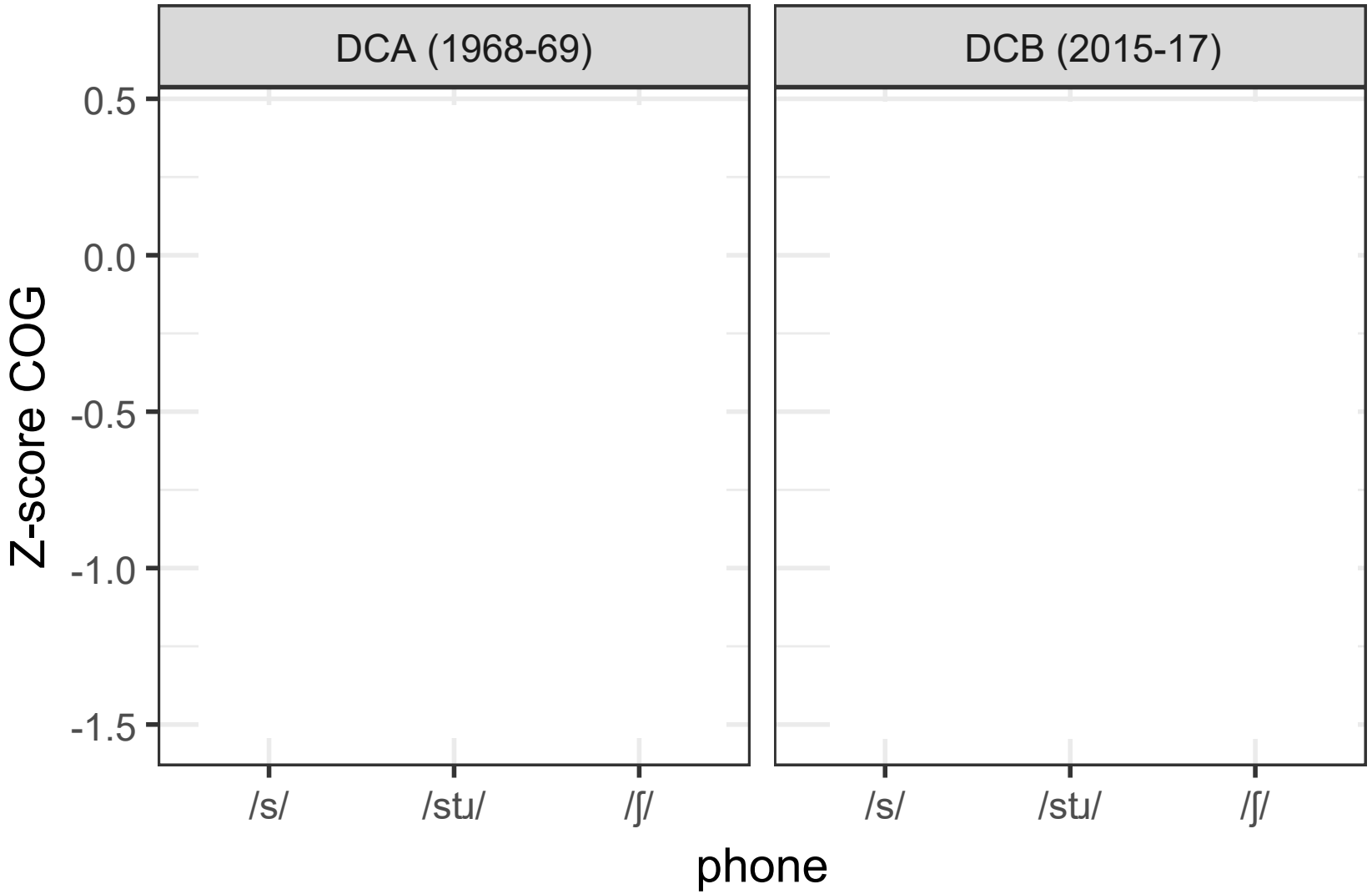
Question 1: /stɹ/ retraction over time

- Dependent variable: **center of gravity** (COG) scaled by speaker
- Independent variables: Three-way interaction between **gender**, **corpus** (DCA vs. DCB), and **phone** (/s/ vs. /stɹ/ vs. /ʃ/). Also included **age** and **log duration**
- Stats: linear-mixed effects model with random intercepts for **speaker** and **word**

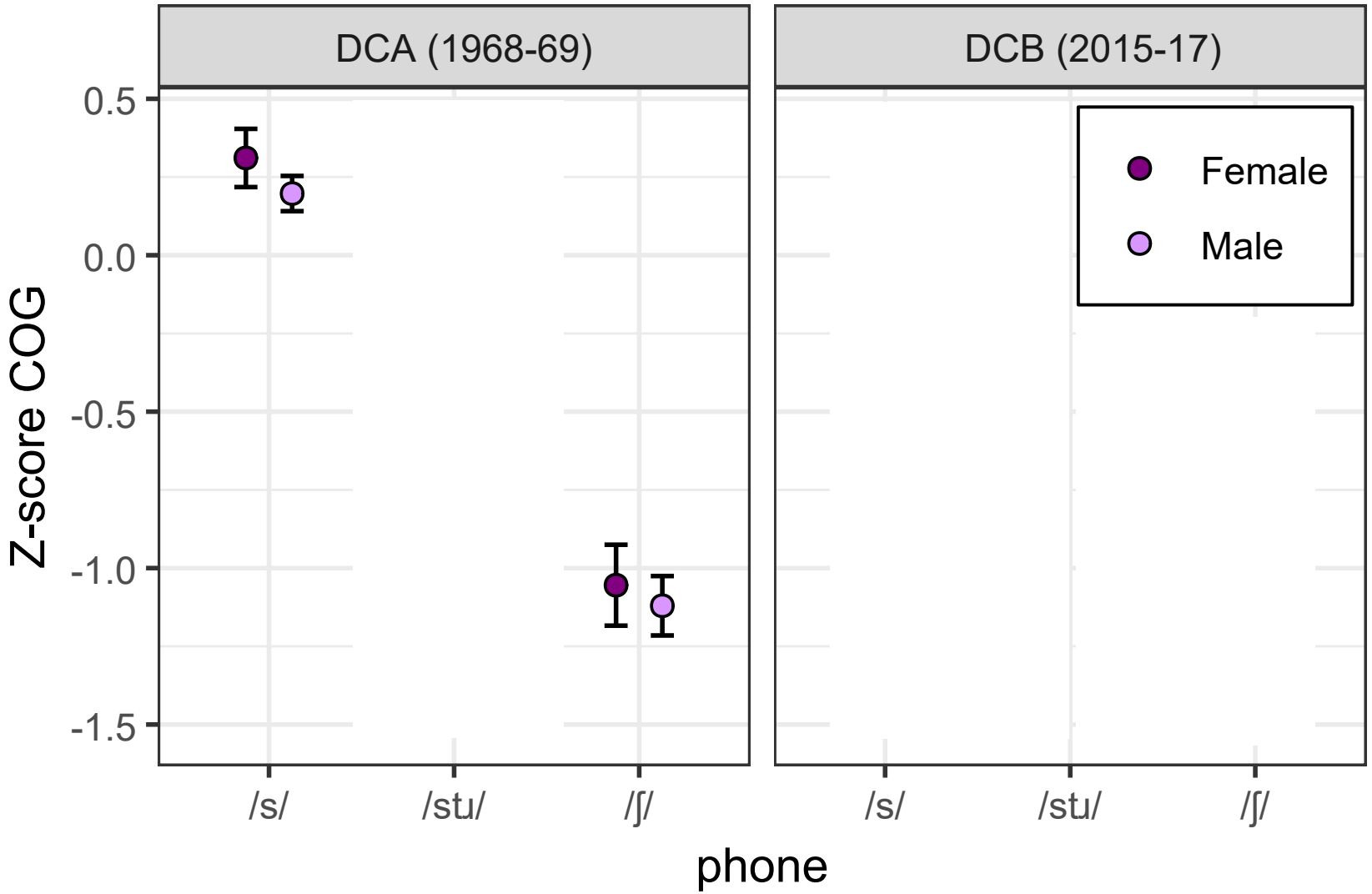
Analysis

Question 2: Coarticulatory bias towards retraction

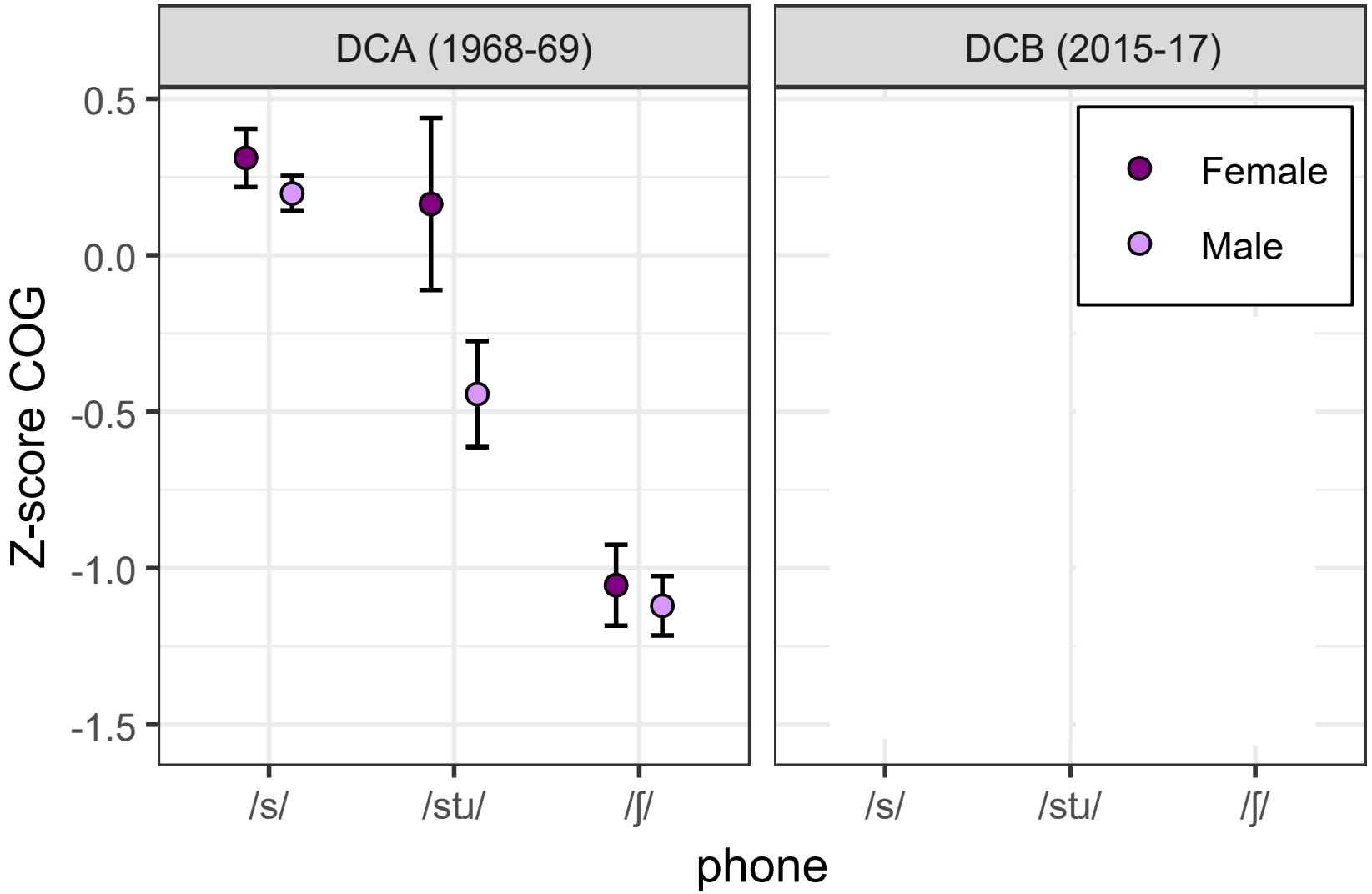
- A speaker's /stɹ/ is retracted if $\geq$ 75.5% of /s/-/ʃ/ COG distance
- **Retraction ratio**: /s/-/stɹ/ distance divided by /s/-/ʃ/ distance
  - Ratio = 0: /stɹ/ and /s/ overlap
  - Ratio = 1: /stɹ/ and /ʃ/ overlap



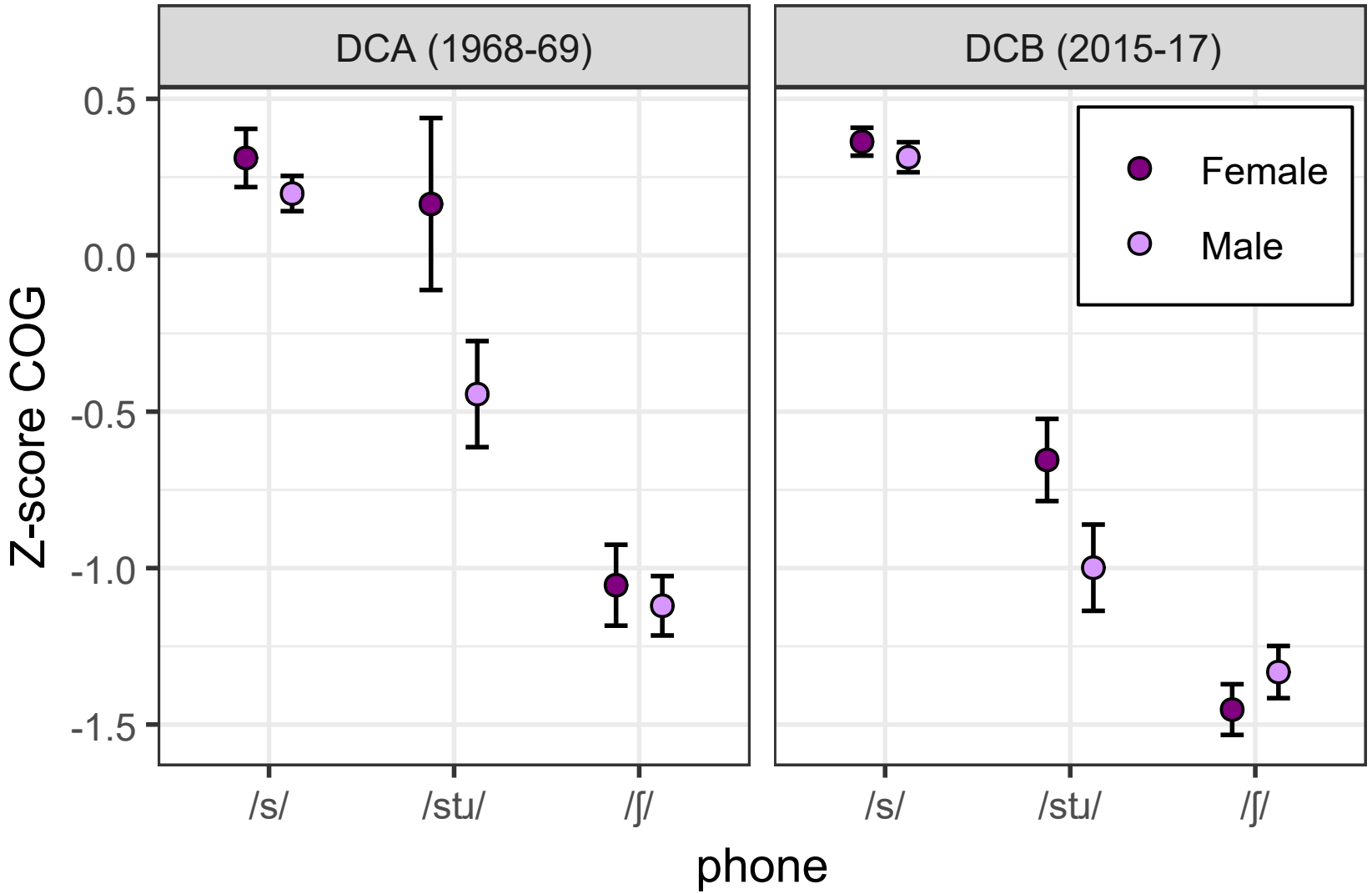
Predicted values from LMER model (R^2=0.397); bars represent 95% confidence intervals



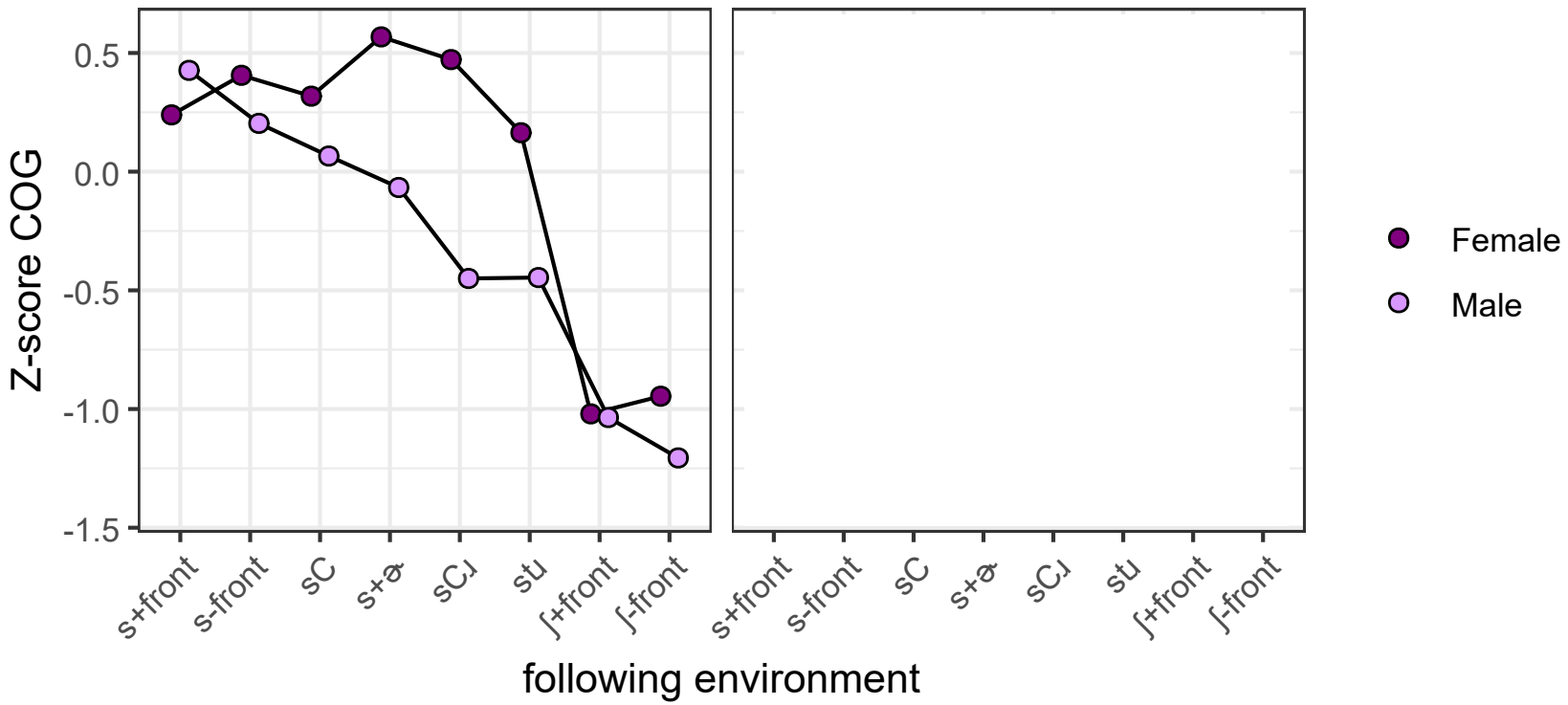
Predicted values from LMER model (R^2=0.397); bars represent 95% confidence intervals



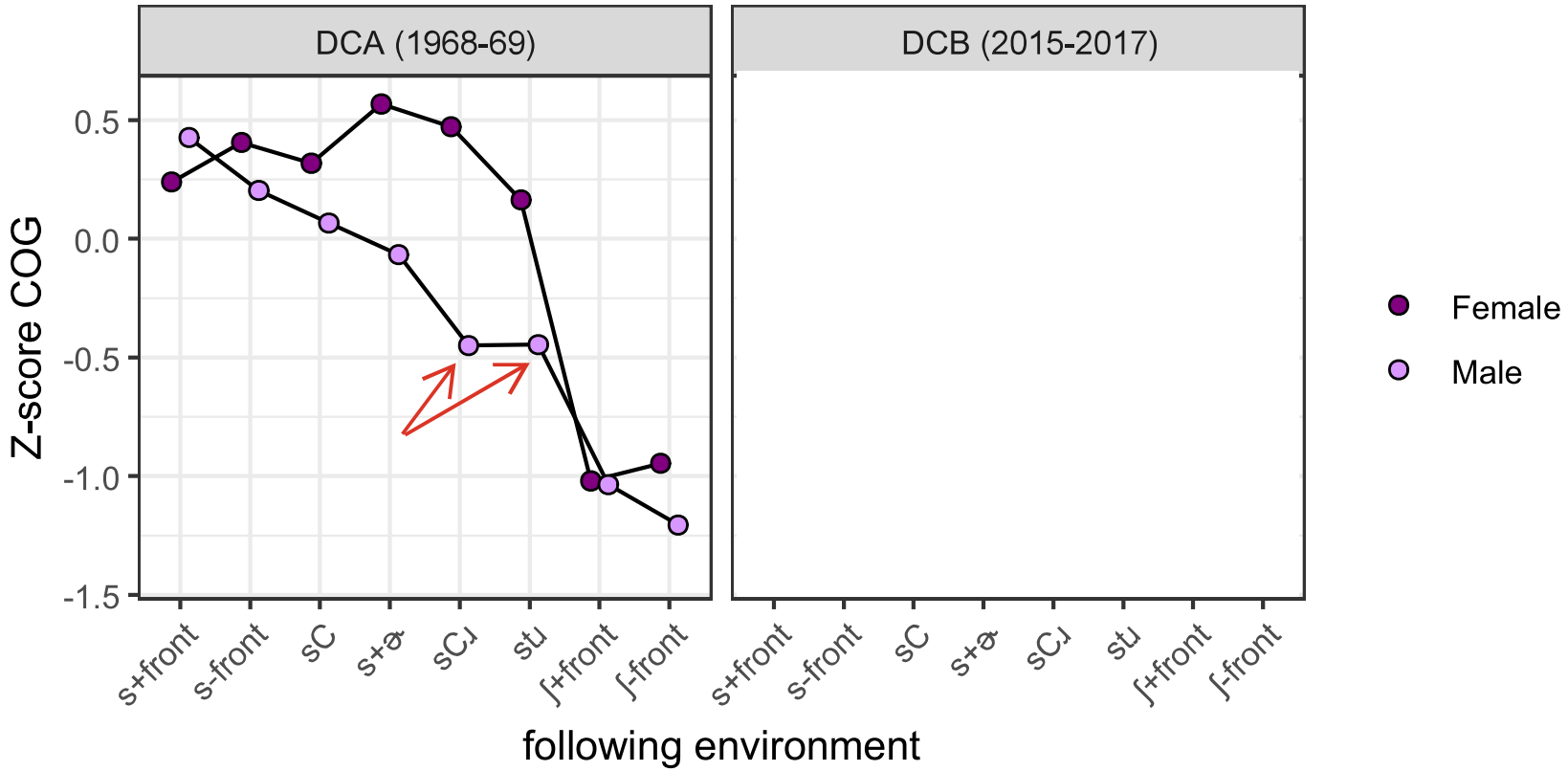
Predicted values from LMER model (R^2=0.397); bars represent 95% confidence intervals



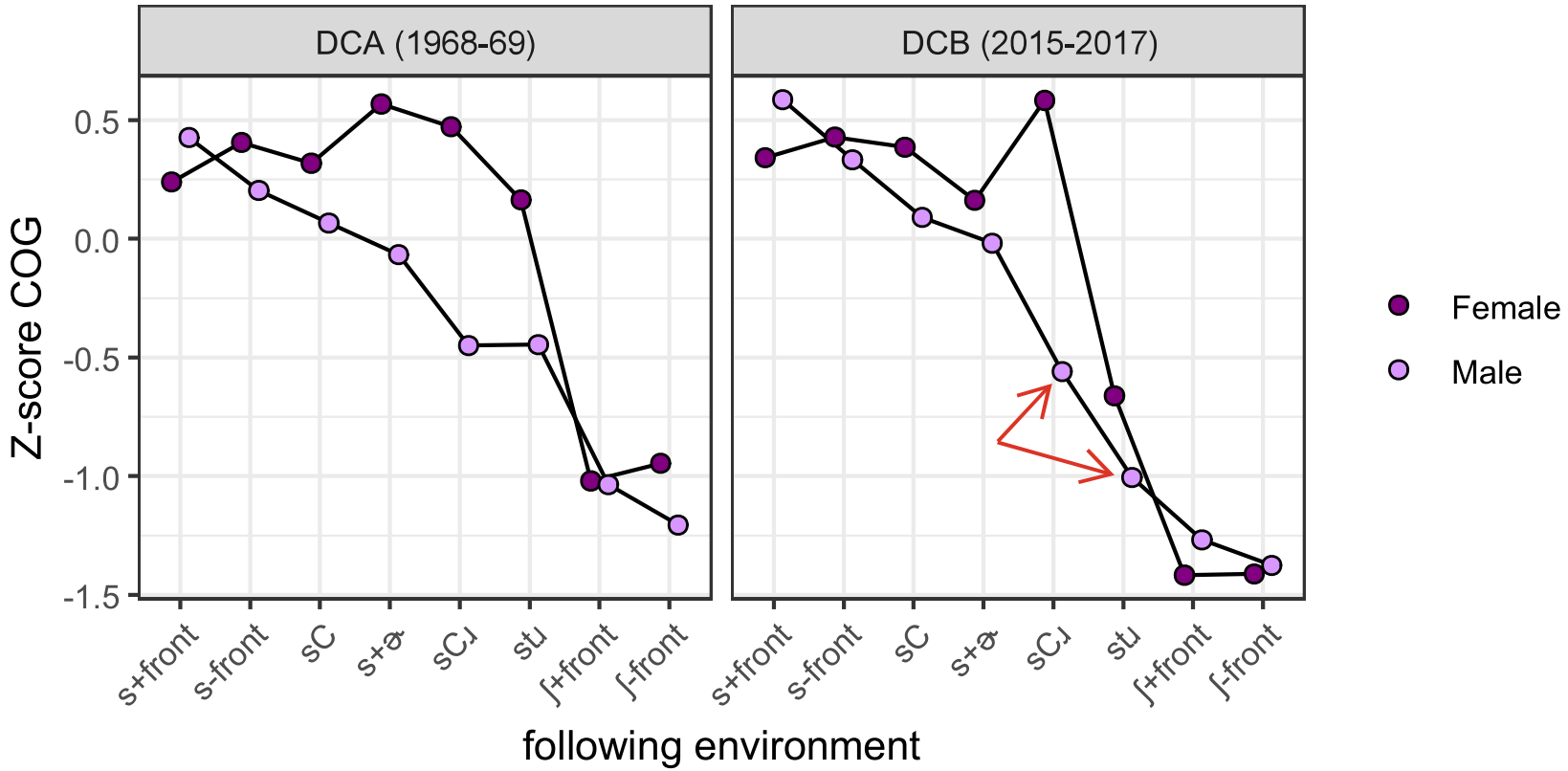
Predicted values from LMER model (R^2=0.397); bars represent 95% confidence intervals



Predicted values from LMER model (R^2=0.406) across following environment

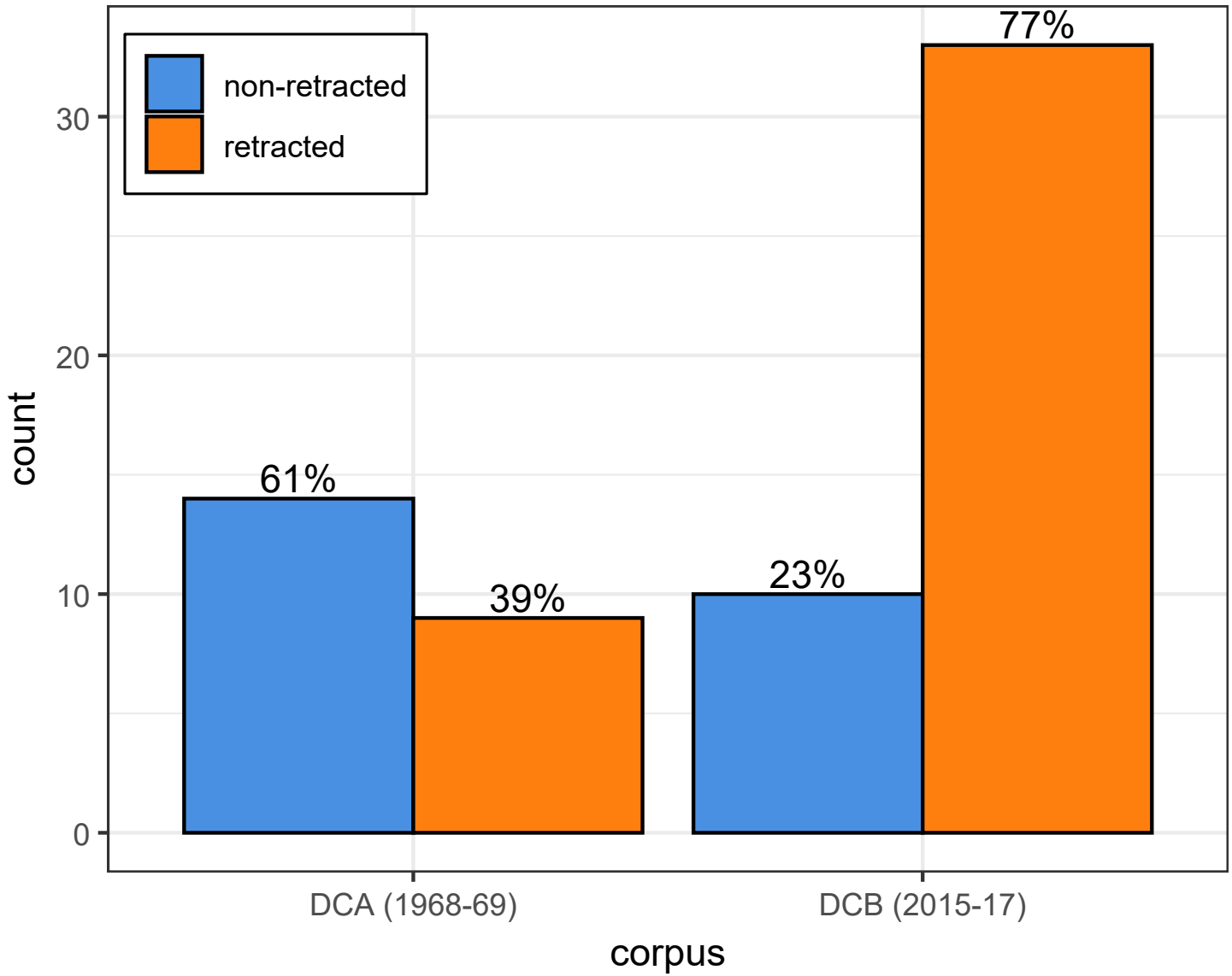


Predicted values from LMER model (R^2=0.406) across following environment

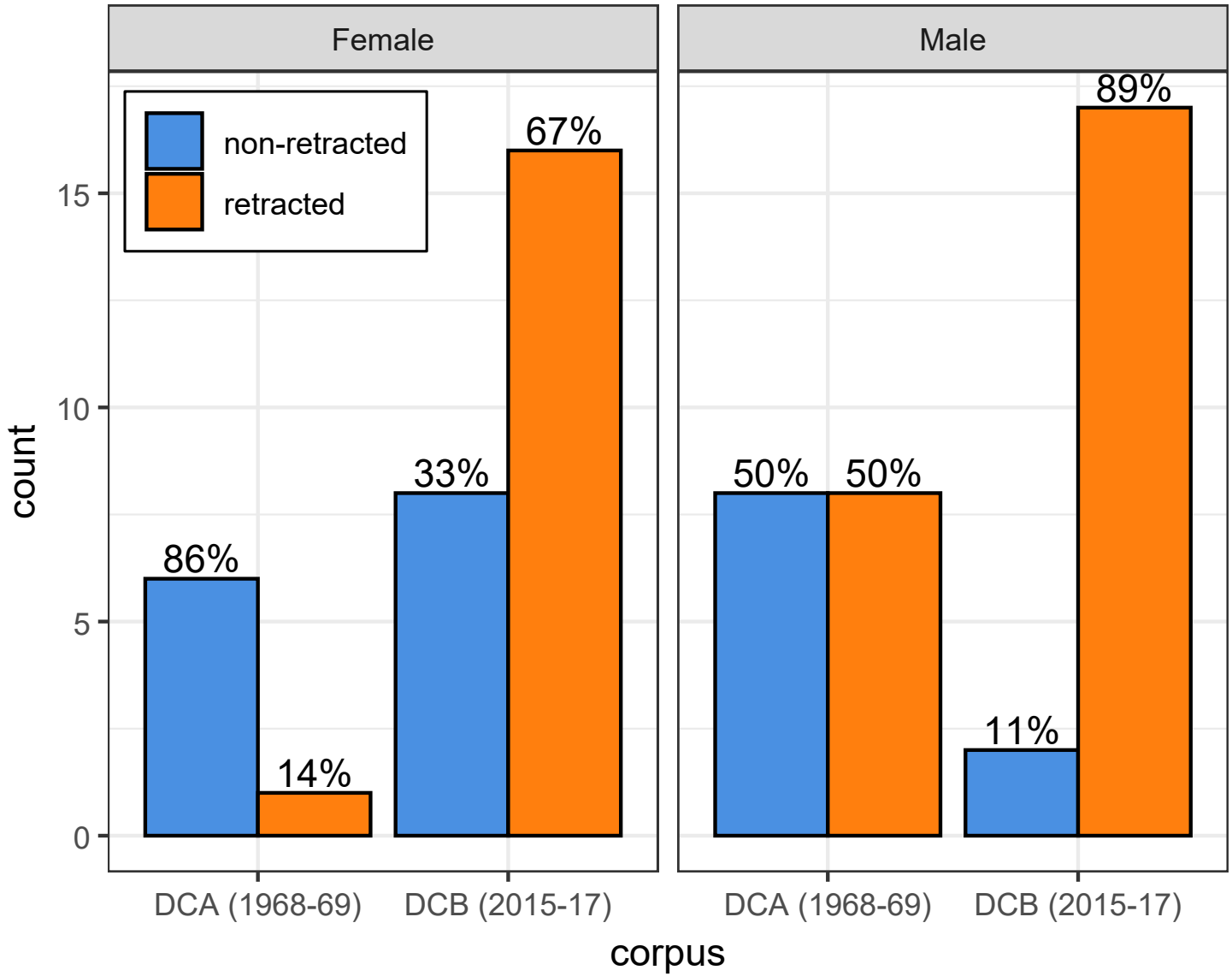


Predicted values from LMER model (R^2=0.406) across following environment

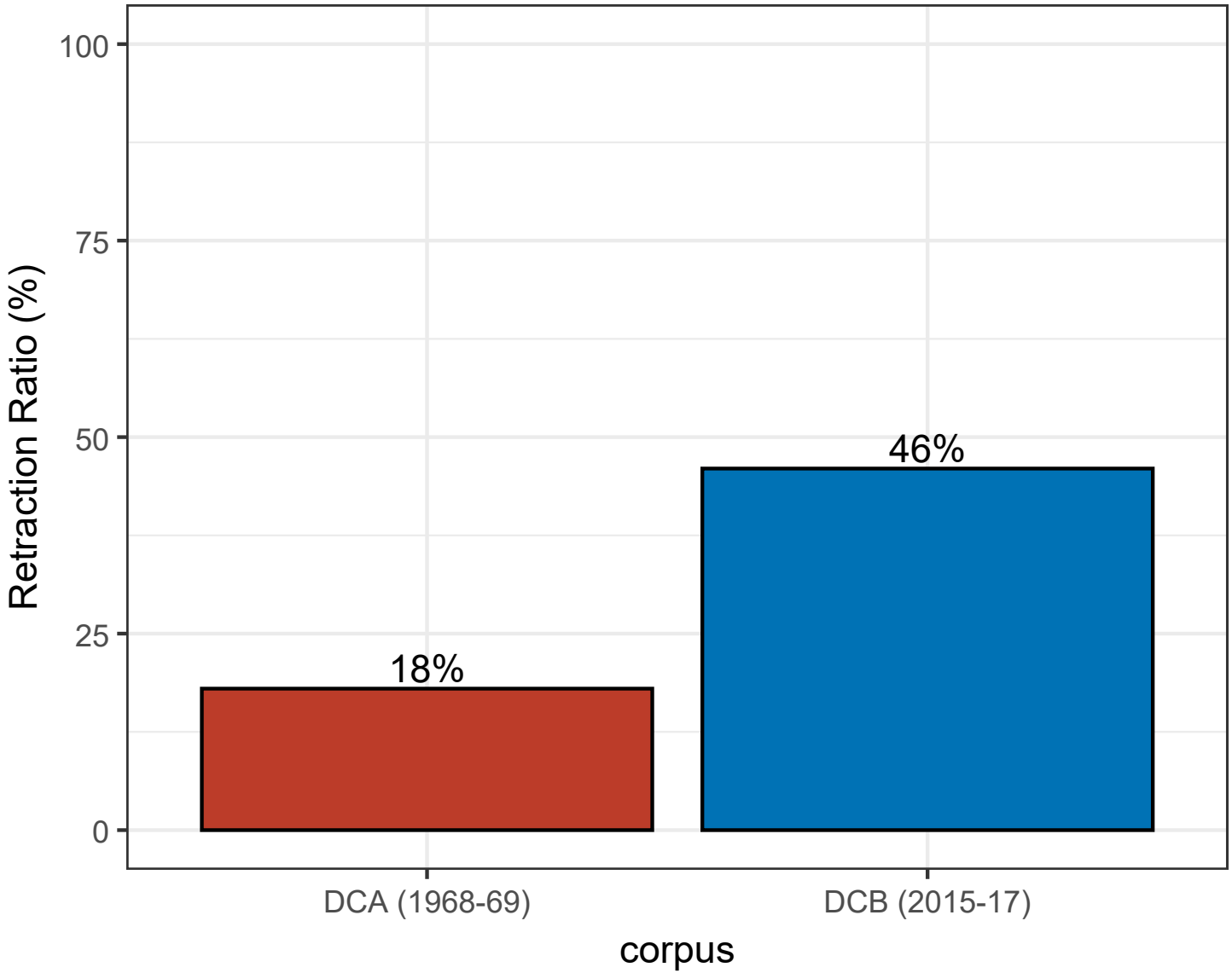
Retracted /stɹ/ across speakers



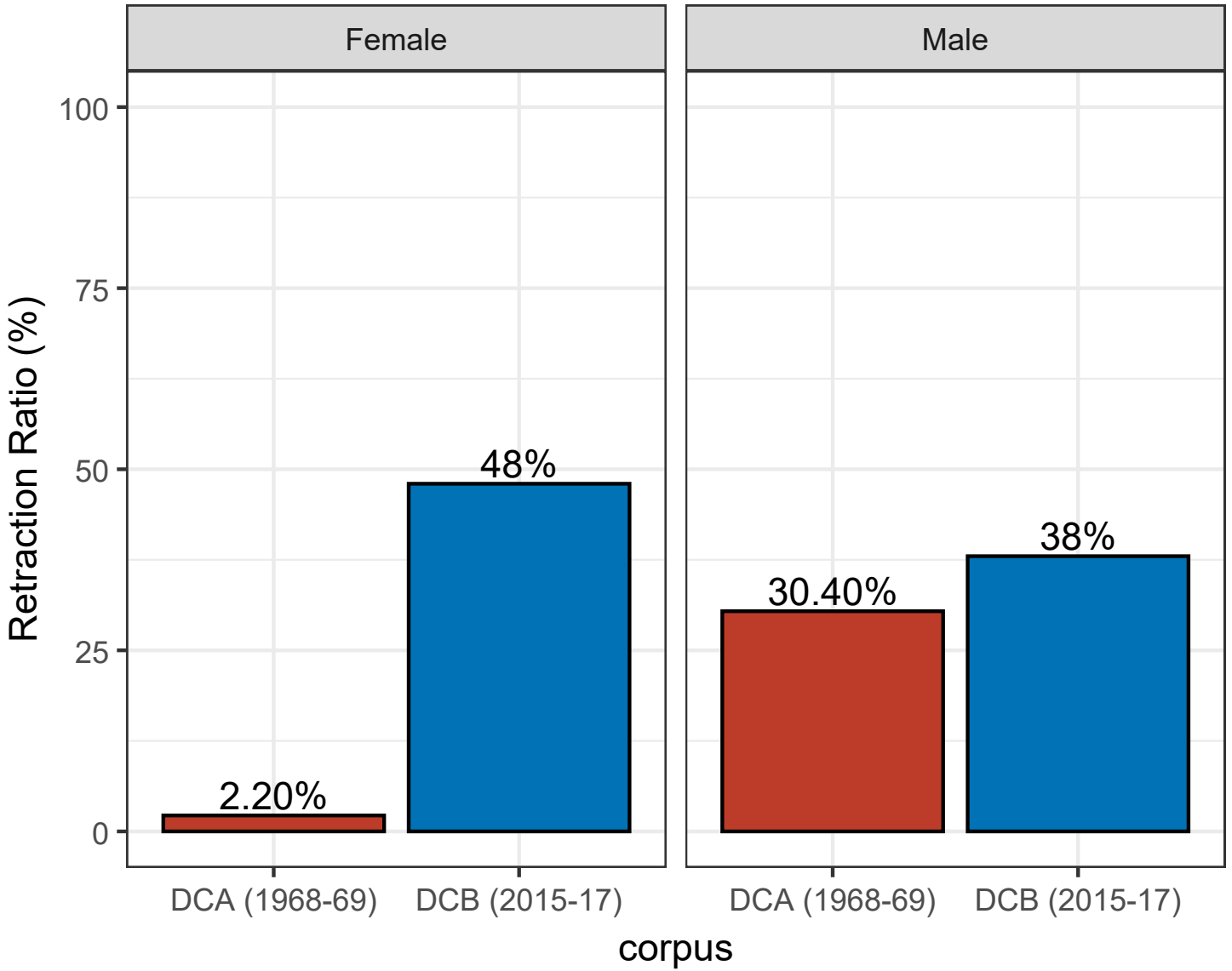
Retracted /stɹ/ across speakers



Coarticulatory bias towards retraction in non-retractors



Coarticulatory bias towards retraction in non-retractors



Changes in the sibilant space

- 89% of retractors in DCA (1968-69) had mean /stɪ/ values *below* their mean /ʃ/

Changes in the sibilant space

- 89% of retractors in DCA (1968-69) had mean /stɪ/ values *below* their mean /ʃ/
- 66% of retractors in DCB (2015-17) had mean /stɪ/ values below their mean /ʃ/

Changes in the sibilant space

- 89% of retractors in DCA (1968-69) had mean /stɪ/ values *below* their mean /ʃ/
- 66% of retractors in DCB (2015-17) had mean /stɪ/ values below their mean /ʃ/
- Interestingly, all retractors had higher COG for /ʃ/ relative to non-retractors

Changes in the sibilant space

- 89% of retractors in DCA (1968-69) had mean /stɪ/ values *below* their mean /ʃ/
- 66% of retractors in DCB (2015-17) had mean /stɪ/ values below their mean /ʃ/
- Interestingly, all retractors had higher COG for /ʃ/ relative to non-retractors
  - When the sibilant space is defined with /s/-/ʃ/ boundaries, non-retractors have wider spaces

Discussion

- /stɹ/ retraction was *not* at the advanced stage in the late 1960s that it was in the 2010s (Gunter et al., 2021)

Discussion

- /stɹ/ retraction was *not* at the advanced stage in the late 1960s that it was in the 2010s (Gunter et al., 2021)
  - Not indexing an older vernacular form

Discussion

- /stɹ/ retraction was *not* at the advanced stage in the late 1960s that it was in the 2010s (Gunter et al., 2021)
  - Not indexing an older vernacular form
- Instead, /stɹ/ retraction increased over time in Washington D.C. AAL

Discussion

- /stɪ/ retraction was *not* at the advanced stage in the late 1960s that it was in the 2010s (Gunter et al., 2021)
  - Not indexing an older vernacular form
- Instead, /stɪ/ retraction increased over time in Washington D.C. AAL
- Males appeared to lead the change, align with pattern found by Stuart-Smith (2020)

Discussion

- /stɪ/ retraction was *not* at the advanced stage in the late 1960s that it was in the 2010s (Gunter et al., 2021)
  - Not indexing an older vernacular form
- Instead, /stɪ/ retraction increased over time in Washington D.C. AAL
- Males appeared to lead the change, align with pattern found by Stuart-Smith (2020)
  - In the 1960s, most women did not retract /stɪ/

Discussion

- /stɪ/ retraction was *not* at the advanced stage in the late 1960s that it was in the 2010s (Gunter et al., 2021)
  - Not indexing an older vernacular form
- Instead, /stɪ/ retraction increased over time in Washington D.C. AAL
- Males appeared to lead the change, align with pattern found by Stuart-Smith (2020)
  - In the 1960s, most women did not retract /stɪ/
  - In the 2010s, most women did

Discussion

- In line with Gunter et al. (2021), many speakers exhibited mean /stɹ/ values below their mean /ʃ/

Discussion

- In line with Gunter et al. (2021), many speakers exhibited mean /stɹ/ values below their mean /ʃ/
  - Not the case with other work on /stɹ/ retraction (Baker et al., 2011; Wilbanks, 2016)

Discussion

- In line with Gunter et al. (2021), many speakers exhibited mean /stɹ/ values below their mean /ʃ/
  - Not the case with other work on /stɹ/ retraction (Baker et al., 2011; Wilbanks, 2016)
- It's possible that this degree of /stɹ/ retraction is dialect-specific

Discussion

- In line with Gunter et al. (2021), many speakers exhibited mean /stɹ/ values below their mean /ʃ/
  - Not the case with other work on /stɹ/ retraction (Baker et al., 2011; Wilbanks, 2016)
- It's possible that this degree of /stɹ/ retraction is dialect-specific
  - Consider the realization of /stɹ/ as [ʃkɹ] in certain varieties of AAL (Fasold and Wolfram, 1972; Thomas, 2007)

Discussion

- The retraction ratio for non-retractors also increased over time

Discussion

- The retraction ratio for non-retractors also increased over time
  - As retracted /stɹ/ becomes more common phonologically, phonetic bias towards retraction also increases

Discussion

- The retraction ratio for non-retractors also increased over time
  - As retracted /stɹ/ becomes more common phonologically, phonetic bias towards retraction also increases
- Generally, non-retractors have wider sibilant spaces than retractors

Discussion

- The retraction ratio for non-retractors also increased over time
  - As retracted /stɹ/ becomes more common phonologically, phonetic bias towards retraction also increases
- Generally, non-retractors have wider sibilant spaces than retractors
  - Speakers with more proximal /s/-/ʃ/ are more likely to retract (Stevens et al., 2019)

Acknowledgments

Thank you to Charlie Farrington, Tyler Kendall, and the entire CORAAL team for making the data so accessible

Baker, Adam, Diana Archangeli, and Jeff Mielke. 2011. Variability in American English s-retraction suggests a solution to the actuation problem. *Language variation and change* 23:347–374.

Farrington, Charlie, and Tyler Kendall. 2019. The Corpus of Regional African American Language: MFA-Aligned. Version 2019.06. *Eugene, OR: The Online Resources for African American Language Project* URL `http://lingtools.uoregon.edu/coraal/aligned/`.

Farrington, Charlie, and Natalie Schilling. 2019. Contextualizing the corpus of regional African American Language, DC: AAL in the nation's capital. *American Speech: A Quarterly of Linguistic Usage* 94:21–35.

Fasold, Ralph W, and Walt Wolfram. 1972. Some linguistic features of Negro dialect. *Language, Speech, and Hearing Services in Schools* 3:16–49.

Gunter, Kaylynn, Charlotte Vaughn, and Tyler Kendall. 2021. Contextualizing /s/ retraction: Sibilant variation and change in Washington DC African American Language. *Language Variation and Change* 33:331–357.

Mielke, Jeff, and Eric Wilbanks. 2015. One Script to Rule Them All (Software). *NCSU Phonetics Lab* .

Reidy, Patrick F. 2013. spectRum. URL `https://github.com/patrickreidy/spectRum`, R package.

Stevens, Mary, Jonathan Harrington, and Florian Schiel. 2019. Associating the origin and spread of sound change using agent-based modelling applied to/s/-retraction in english. *Glossa: a journal of general linguistics* .

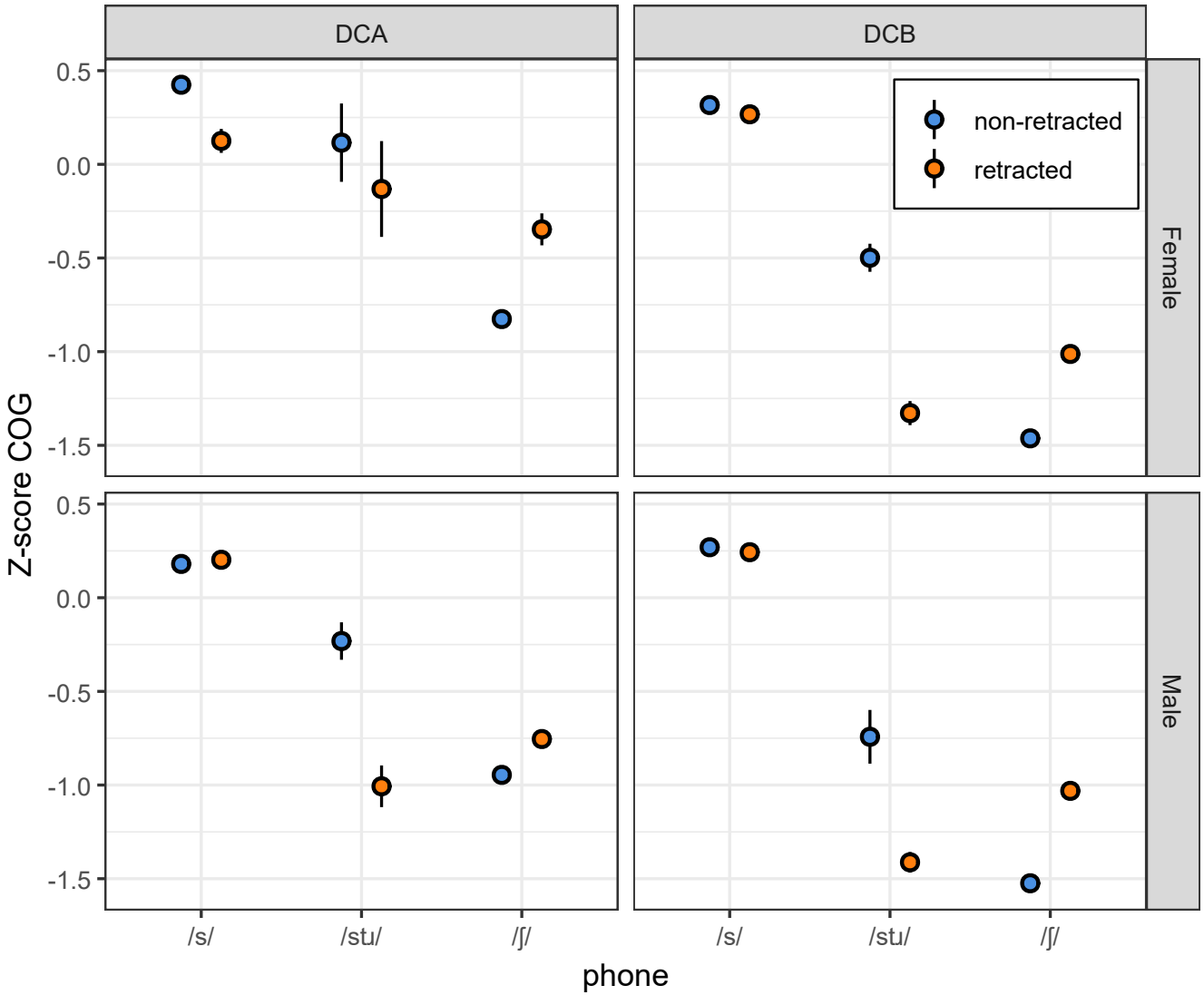
Stuart-Smith, Jane. 2020. Changing perspectives on /s/ and gender over time in Glasgow. *Linguistics Vanguard* 6.

Stuart-Smith, Jane, Morgan Sonderegger, Rachel Macdonald, Jeff Mielke, Michael McAuliffe, and Erik R. Thomas. 2019. Large-scale acoustic analysis of dialectal and social factors in English /s/-retraction. In *Proceedings of ICPhS 2019*, 1273–1277. .

Thomas, Erik R. 2007. Phonological and phonetic characteristics of African American vernacular English. *Language and Linguistics Compass* 1:450–475.

Wilbanks, Eric. 2016. Social and structural constraints on a phonetically-motivated change in progress:(str) retraction in Raleigh, NC. *UC Berkeley PhonLab Annual Report* 12.

Bonus Graph


DCA
DCB
Female
Male
non-retracted
retracted
Z-score COG
0.5
0.0
-0.5
-1.0
-1.5
/s/
/stɹ/
/ʃ/
phone